Fiscal Year:	FY 2006	**Task Last Updated:**	FY 12/21/2006
PI Name:	Sayler, Gary Ph.D.		
Project Title:	Prototype Biophotonic Biosensor for Monitoring VOC Contaminants in Spacecraft Habitats		
Division Name:	Human Research		
Program/Discipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES		
Program/Discipline--Element/Subdiscipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES--Advanced environmental monitoring and control		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	sayler@utk.edu	**Fax:**	FY 865-974-8086
PI Organization Type:	UNIVERSITY	**Phone:**	865-974-8080
Organization Name:	University of Tennessee		
PI Address 1:	676 Dabney Hall		
PI Address 2:			
PI Web Page:	http://www.ceb.utk.edu		
City:	Knoxville	**State:**	TN
Zip Code:	37996	**Congressional District:**	2
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	03-OBPR-01
Start Date:	06/30/2004	**End Date:**	09/30/2006
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Ripp, Steven (University of Tennessee) Islam, Syed (University of Tennessee) Blalock, Benjamin (University of Tennessee) Garland, Jay (Dynamac Corp.) Del Castillo, Linda (Jet Propulsion Laboratory)		
Grant/Contract No.:	NNJ04HF02A		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	This research and development effort seeks to extend whole cell biosensor systems to applications in continuous monitoring of high priority toxic, volatile organic chemicals (VOCs) in the spacecraft environment. Since our first report of bioluminescent bioreporters functioning as autonomous biosensors for chemical agents in environmental samples (1), in less than a decade whole-cell sensing systems have become widely recognized for their robust, sensitive, and specific recognition of target analytes. The practical demonstration that engineered bacterial strains could be mated directly to a microluminometer chip to create a bioluminescent bioreporter integrated circuit (BBIC) has stimulated additional designs for BBIC based remote sensors capable of target analyte detection in various gas-liquid and solid environmental samples with sensitivity and reproducibility rivaling conventional analytical methods (2). With continuing advances in the ability to engineer even more novel sensing strategies into living cells, using approaches such as DNA-shuffling or computational design, it is clear that smart, robust whole cell biosensor systems can be developed for functional deployment in monitoring air quality characteristics in the spacecraft environment for long-term protection of astronaut health. To realize this true potential, appropriate bioreporter sensor strains must be developed and incorporated in a BBIC sensor with optimal integrated circuit technology for signal detection, processing, and communication in a 'ruggedized' format capable of sustaining the bioreporter for extended analytical timeframes. This research seeks to develop specific bioreporter strains to meet NASA defined monitoring needs of priority VOCs and to create a BBIC based prototype biosensor for ground based verification and in flight testing in support of NASA's advanced life support system programs.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The detection of chemicals using miniaturized autonomous biosensors can be applied in office buildings, homes, and other enclosed structures (submarines, for example) to continuously monitor air quality (i.e., mold growth or other toxic buildups associated with sick building syndromes) and for the detection of chemical warfare agents.
Task Progress:	The bioluminescent bioreporter integrated circuit (BBIC) is composed of whole-cell reporter organisms that rapidly (= 1 hour), continuously, specifically, and reproducibly generate bioluminescent light in response to part-per-billion (ppb) concentrations of target chemicals, and a silicon-based microluminometer that detects and processes the light signals. BBIC sensors function as a network of autonomous detectors capable of continuous on-line, real time monitoring in a low-power, low-mass, low-maintenance, redundant framework. Compatible with cabled or wireless interfaces, BBICs provide a systems integrated approach for wide array biosensing of the chemical environment. Our goals in this research effort were to genetically engineer bioluminescent bioreporter organisms capable of specifically detecting the NASA priority 1 volatile organic chemicals (VOCs) dichloromethane and formaldehyde and test their response kinetics on an upgraded and ruggedized BBIC microelectronic platform. As well, a controlled atmosphere test chamber was designed for BBIC testing purposes. This will serve to verify that bioreporter sensing of atmospheric chemical contaminants on a microchip platform is feasible. We have constructed a luxCDABE-based bioluminescent bioreporter for dichloromethane. This bioreporter was capable of detecting dichloromethane under both static and flow-through conditions within a four hour response time. Exposure to other relevant volatiles such as chloroform, carbon tetrachloride, formaldehyde, trichloroethlyene, and tetrachloroethylene did not induce bioluminescence from the bioreporter. A bioluminescent bioreporter for formaldehyde is also under construction. The BBIC microluminometer chip was upgraded from a previous version to improve ruggedness and reliability. These upgrades included: - A larger photodiode area, thus, with the same bioluminescent input, the chip will process faster and shorter output pulses - An on-chip 1.2V band-gap voltage reference to reduce the number of off-chip components - An on-chip 16-bit counter to generate a 16-bit parallel output that continuously displays the pulse width of the BBIC output. This enables a 4-bit (Hex) 7-segment LED to be connected to the 16-bit parallel output of the BBIC chip to create a portable BBIC circuit that displays the pulse width of the BBIC output with 16-bit binary resolution or 4-bit hex resolution. - A USB interface The BBIC chip was also integrated into an existing handheld probe that we have constructed for remote chemical detection. This device allows for user defined plug-and-play monitoring of desired chemical agents via selection and insertion of appropriate bioreporter organisms. Bioluminescence in response to target chemical exposure is detected by the BBIC and results downloaded via the USB interface. These tasks have resulted in final deliverables consisting of: • A bioluminescent bioreporter capable of detecting dichloromethane. Comprehensive BBIC response profiling still needs to be established via enclosed chamber testing. • A bioluminescent bioreporter capable of detecting formaldehyde. Although still under construction, it will be completed and preliminarily tested within the next three months. • BBIC biochip with following upgrades: - Larger photodiode - On-chip 1.2V band-gap voltage reference - On-chip 16-bit counter - Ruggedized packaging - USB interface - Integration into a handheld BBIC probe
Bibliography Type:	Description: (Last Updated: 03/16/2017)
Articles in Other Journals or Periodicals	Ripp S. "Biosensor monitoring of the cleanroom environment." CleanRooms Magazine. 2006 Apr;20:25-9. , Apr-2006

Articles in Peer-reviewed Journals	Islam SK, Vijayaraghavn R, Zhang M, Ripp S, Caylor S, Weathers B, Moser S, Terry S, Blalock B, Sayler GS. "Integrated circuit biosensors using living whole-cell bioreporters." IEEE Transactions on Circuits and Systems, in press, January 2007. http://dx.doi.org/10.1109/TCSI.2006.887982 , Jan-2007
Articles in Peer-reviewed Journals	Vijayaraghavn R, Islam SK, Zhang M, Ripp S, Caylor S, Weathers B, Moser S, Terry S, Blalock B, Sayler GS. "A bioreporter bioluminescent integrated circuit for very low-level chemical sensing in both gas and liquid environments." Sensors and Actuators B: Chemical, In Press, Corrected Proof, Available online 29 November 2006. http://dx.doi.org/10.1016/j.snb.2006.10.064 , Nov-2006
Articles in Peer-reviewed Journals	Sayler GS, Simpson ML, Cox CD. "Emerging foundations: nano-engineering and bio-microelectronics for environmental biotechnology." Curr Opin Microbiol. 2004 Jun;7(3):267-73. Review. PMID: 15196494 , Jun-2004
Books/Book Chapters	Ripp S, Garland J, Blalock BJ, Islam SK, Sayler GS. "Bioluminescent bioreporter integrated circuit sensing of the chemical and biological spacecraft environment." in "Bioluminescence and Chemiluminescence Progress and Perspectives." Ed. A. Tsuji, M. Matsumoto, M. Maeda, L. J. Kricka, and P. E. Stanley. Hackensack, New Jersey : World Scientific, 2005., Jul-2005
Papers from Meeting Proceedings	Ripp S, Moser S, Weathers B, Caylor S, Brigati JR, Blalock B, Islam SK, Sayler G. "Bioluminescent bioreporter integrated circuit (BBIC) sensing of the chemical and biological environment." The American Society for Microbiology Biodefense Conference, Washington, DC, Feb. 15-18, 2006. The American Society for Microbiology Biodefense Conference, 2006. , Feb-2006
Papers from Meeting Proceedings	McReynolds WM, Ripp S, Sayler GS. "Environmental monitoring for dichloromethane using lux based bioluminescent bioreporters." Annual Meeting of the American Society for Microbiology, Orlando, FL, May 21-25, 2006. Annual Meeting of the American Society for Microbiology, May 2006. , May-2006
Papers from Meeting Proceedings	Ripp S, Moser S, Weathers B, Blalock B, Islam S, Sayler G. "Bioluminescent bioreporter integrated circuits (BBICs)." Biosensors Congress, Toronto, Canada, May 10-12, 2006. Biosensors Congress, May, 2006. , May-2006
Papers from Meeting Proceedings	Ripp S, Moser S, Brigati JR, Weathers B, Caylor S, Blalock B, Islam SK, Sayler G. "Bioluminescent bioreporter integrated circuit (BBIC) sensors." American Society for Microbiology BioMicroNano conference, San Francisco, CA, Jan. 15-18, 2006. American Society for Microbiology BioMicroNano conference, Jan. 2006. , Jan-2006